

SEIP assignment assessment

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Successful build-and-push pipeline completion.

The screenshot shows a GitHub Actions workflow run for the 'build-and-push' job. The job is completed successfully, indicated by a green checkmark and the text 'succeeded 3 minutes ago in 18s'. The workflow steps are listed on the right, showing a sequence of actions like 'Set up job', 'Check out repository code', 'Authenticate to GHCR', 'Build Docker image', 'Push Docker image', 'Post Authenticate to GHCR', 'Post Check out repository code', and 'Complete job'. Each step has a duration listed on the right.

Step	Duration
Set up job	1s
Check out repository code	1s
Authenticate to GHCR	0s
Build Docker image	8s
Push Docker image	5s
Post Authenticate to GHCR	0s
Post Check out repository code	0s
Complete job	0s

Cluster State proof.

```
● ariseconomidis@Ariss-MBP-2 seip_assignment_1_2026 % kubectl get all -n default
NAME                                     READY   STATUS    RESTARTS   AGE
pod/echo-api-deployment-6c64b4ff8d-j4frs 1/1     Running   1 (2d17h ago)  2d17h
pod/echo-api-deployment-6c64b4ff8d-jppkw 1/1     Running   1 (2d17h ago)  2d17h
pod/echo-api-deployment-6c64b4ff8d-rz8rd 1/1     Running   1 (2d17h ago)  2d17h

NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
service/echo-api-service            ClusterIP     10.109.144.138  <none>        80/TCP     2d17h
service/kubernetes                  ClusterIP     10.96.0.1       <none>        443/TCP    2d17h

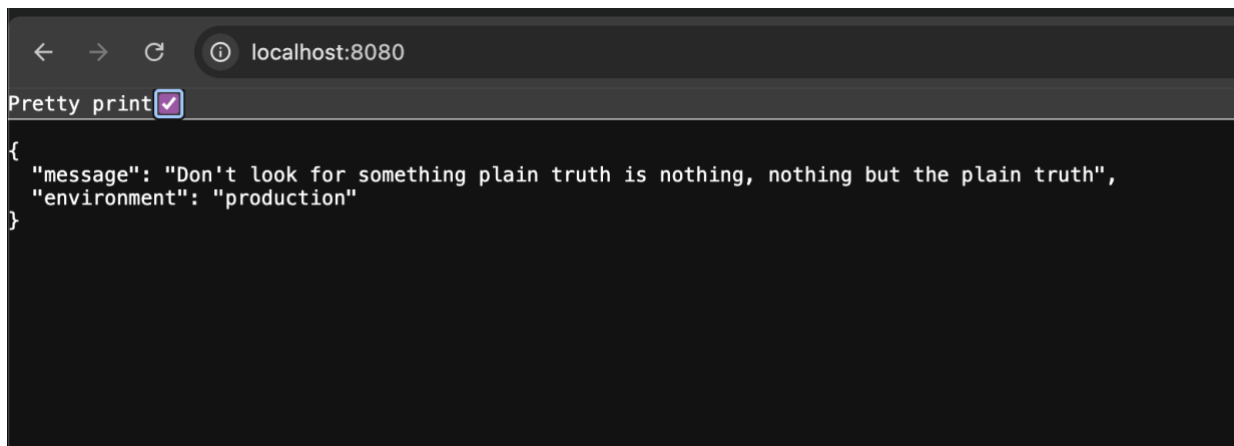
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/echo-api-deployment 3/3     3             3           2d17h

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/echo-api-deployment-6c64b4ff8d 3         3         3       2d17h
replicaset.apps/echo-api-deployment-6dd945ddd 0         0         0       2d17h
● ariseconomidis@Ariss-MBP-2 seip_assignment_1_2026 %

● ariseconomidis@Ariss-MBP-2 seip_assignment_1_2026 % kubectl get configmap,secret
NAME                                DATA   AGE
configmap/echo-api-config           2       2d17h
configmap/kube-root-ca.crt          1       2d18h

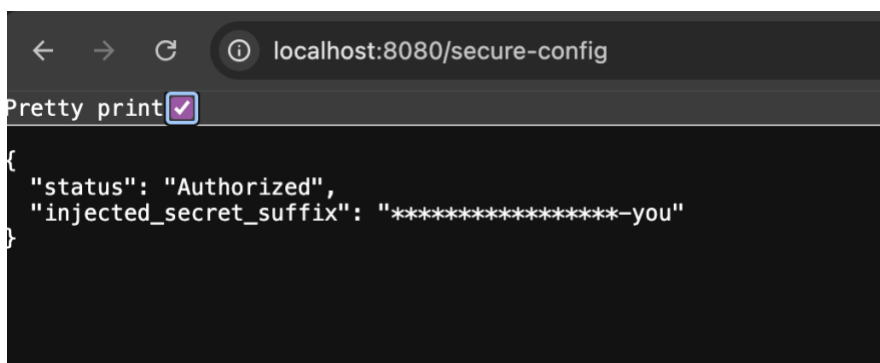
NAME                                TYPE    DATA   AGE
secret/echo-api-secret              Opaque  1       2d17h
● ariseconomidis@Ariss-MBP-2 seip_assignment_1_2026 %
```

Application Verification Proof



A screenshot of a web browser window with the address bar showing 'localhost:8080'. The page content displays a JSON object with two fields: 'message' and 'environment'. The 'message' field contains the text 'Don't look for something plain truth is nothing, nothing but the plain truth', and the 'environment' field contains the value 'production'. A 'Pretty print' button with a checkmark icon is visible at the top left of the content area.

```
{
  "message": "Don't look for something plain truth is nothing, nothing but the plain truth",
  "environment": "production"
}
```



A screenshot of a web browser window with the address bar showing 'localhost:8080/secure-config'. The page content displays a JSON object with two fields: 'status' and 'injected_secret_suffix'. The 'status' field contains the value 'Authorized', and the 'injected_secret_suffix' field contains the value '*****-you'. A 'Pretty print' button with a checkmark icon is visible at the top left of the content area.

```
{
  "status": "Authorized",
  "injected_secret_suffix": "*****-you"
}
```

Regarding AI usage

Generative AI models (ChatGPT, Claude) were utilised throughout the completion of this assignment. Those models were mainly used under the following scenarios:

- Explaining the way technologies ought to be used (docker, kubernetes, github actions, etc) and providing examples.
- Giving suggestions regarding the debugging of modules that do not seem to behave accordingly.
- Providing the exact syntax on certain configurations (e.g. run module in Dockerfile), as well as the proper terminal commands for every step of the deployment process.
- Helping with the contents and styling of Assignment.md,

the most useful of which would most likely be helping me understand the inner workings of the kubernetes mechanism and how the different .yaml files interact with the javascript file. Debugging was also proven to be much easier with the use of LLM's.

Nevertheless, some suggestions I received from chatbots caused issues. As shown in my repository, the first time I ran the build-and-push GitHub Actions job, the Docker image failed to build because its name contained uppercase characters. This happened because of a suggestion to use my GitHub username when tagging the image. Since Docker image names must be lowercase, and my GitHub username contains uppercase letters, the process failed.

Future Outlook

Under a realistic development scenario, the application would need to be reached through a proper hostname and routing layer. So I would replace the local `kubectl port-forward` with an Ingress controller, like `traefik` (although I am not exactly sure if it is provided for free or not), to manage external access. When it comes to scaling up the project, tools like Prometheus and Grafana would be extremely useful in monitoring the health of the pods and extracting metrics regarding the use of resources, request traffic, and potential failures of the system. Finally, on the CI/CD side, I would try to improve security by adding image vulnerability scanning before pushing to GHCR, preventing unsafe images from being deployed. This essentially means interjecting between the stages of building and pushing an image. Naturally, there is an already implemented tool called Trivy that scans for vulnerabilities and labels images on a severity of threat level (LOW, MEDIUM, HIGH, CRITICAL) and blocks them from being deployed accordingly.